

Kit SDS Cover Sheet

Document ID: OSR6111-75: Version 13
Revision Date (year/month/day) 2025/06/13
Last Revision Date (year/month/day) 2024/06/18

Product information

Product name Direct Bilirubin
Part number OSR6211, OSR6111, OSR6611
Additional product information
For Professional Use Only

Components

Description Direct Bilirubin Blank R1
Direct Bilirubin Color R1

Transport information

Shipping information	IATA	IMDG	US DOT	European ADR	Canadian TDG
UN/ID number	3264	3264	3264	3264	3264
UN proper shipping name	Corrosive liquid, acidic, inorganic, n.o.s. (sulphuric acid, hydrochloric acid)				
Transport hazard class(es)	8 Corrosives	8 Corrosive substances	8 Corrosive material	8 Corrosive substances	8 Corrosives
Subsidiary risk	None	None	None	None	None
Classification code	Not applicable	Not applicable	Not applicable	C1	Not applicable
Packing group	II	II	II	II	II
Special provisions	A3, A803	274	386	274	16
Additional information					
IATA ERG code	8L	Not applicable	Not applicable	Not applicable	Not applicable
EmS	Not applicable	F-A, S-B	Not applicable	Not applicable	Not applicable
NAERG code	Not applicable	Not applicable	154	Not applicable	154

Environmental hazards

Marine pollutant	Not applicable	No	Not applicable	Not applicable	Not applicable
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Special precautions for user

Warning: Corrosive substances.



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Section 1 Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier		
Product name	Direct Bilirubin Blank R1	
Part number	Component of P/N OSR6111, OSR6211, OSR6611	
1.2 Relevant identified uses of the substance or mixture and uses advised against		
Product use	For In Vitro Diagnostic Use. See product literature for details. For Professional Use Only	
1.3 Details of the supplier of the safety data sheet		
Manufacturer	Beckman Coulter, Inc. 250 S. Kraemer Blvd Brea, CA 92821, U.S.A. Tel: 800-854-3633	
Supplier	CANADA Beckman Coulter Canada LP 7075 Financial Drive Mississauga, ON L5N 6V8 Canada 1-800-463-7828	UNITED KINGDOM Beckman Coulter (UK) Ltd. Amersham Place Little Chalfont Buckinghamshire United Kingdom, HP7 9NA 01494 441181
	AUSTRALIA Beckman Coulter Australia Pty Ltd 23-27 Chaplin Drive Lane Cove NSW 2066 Australia ABN 81 002 011 672 24 Hour emergency contact phone number: 1800 060 881	SWITZERLAND Beckman Coulter Eurocenter SA 22, rue Juste-Olivier, Case Postale 1044, CH-1260 Nyon 1, Switzerland. Telephone: +41 (0)22 365 36 11 Monday through Friday, 9:00 am to 7:00pm)

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Section 1 Identification of the substance/mixture and of the company/undertaking (Continued)

NEW ZEALAND
Beckman Coulter NZ
Unit J, 33 Walmsley Road, Otahuhu,
Auckland 1062, New Zealand
Hours available: 08:30 - 17:00

Beckman Coulter Ireland Inc.
Lismeehan
O'Callaghan's Mills
Co. Clare
Ireland
Tel: 353 (0)65 6831100

ICELAND / ÍSLAND
Beckman Coulter AB
Ekbacksvägen 28
168 69 Bromma
Sweden
Phone No.: +46 80564 85 900
Hours available: 08.00-16.30

MALTA
DX Distributor:
Cherubino Ltd
DELFI Building, Sliema Road, Gzira,
GZR 1637
Telephone: +356 21343270
Hours available: 08:30 – 17:00

e-mail address

SDSNT@beckman.com
Further information Contact:
Customer support Unit, Beckman Coulter Ireland Inc.
Technical Service Department Tel. +001-800-854-3633 (PST)
E-mail: Techsupportuk@beckman.com

1.4 Emergency telephone number

Telephone number (24H)

Chemtrec Emergency Tel No. U.S.A. 800-424-9300, International (001) 703-527-3887
Tel +353 (0)65 683 1170; 08:00 - 16:30 hrs Mon-Thur, 08:00 - 15:30 hrs Fri (GMT) Tel +001-800-223-0130 (PST)

Distributor and emergency phone no.

Refer to attached list, Document ID: [472050](#), for local distributor and emergency phone numbers.

UNITED STATES - Emergency Phone (24h): Chemtrec (800) 424-9300, International (001) 703-527-3887

CANADA - Poison Centre: 1-844-764-7669; Centre antipoison du Québec: 1-800-463-5060

UNITED KINGDOM - For UK and Scotland: Emergency Call 999

IRELAND - National Poisons Information Centre Phone No.: Members of Public: +353 (01) 809 2166 (8:00 am to 10:00 pm 7 days a week); Healthcare Professionals: +353 (01) 809 2566 (24 hour service)

AUSTRALIA - 24 Hour emergency contact phone number: 1800 060 881

NEW ZEALAND - 24 Hour emergency number: 0800 446 109

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Section 1 Identification of the substance/mixture and of the company/undertaking (Continued)

Section 2 Hazards identification

2.1 Classification of the substance or mixture

Product description

In vitro diagnostic reagent.
Colorless; Liquid; Characteristic odor

Classification according to EC 1272/2008 (CLP/GHS)

Skin Corrosion Category 1, H314
Eye Damage Category 1, H318

Classification according to US-OSHA (HCS 29 CFR 1910.1200) and UN GHS

Skin Corrosion Category 1
Eye Damage Category 1

2.2 Label elements

According to EC 1272/2008 (CLP/GHS), US-OSHA and UN GHS

Hazardous ingredients

Sulfuric Acid
Sulfosalicylic Acid, Dihydrate
Hydrochloric Acid

Pictogram



Signal word

DANGER

Hazard statements

H314 Causes severe skin burns and eye damage.

Precautionary statements

Prevention

P280 Wear protective gloves, protective clothing and eye/face protection.

Response

P301+P330+P331 IF SWALLOWED: rinse mouth. Do NOT induce vomiting.

P303+P361+P353 IF ON SKIN (or hair): Rinse skin with water.

P304+P340 IF INHALED: Remove person to fresh air and keep at rest in a position comfortable for breathing.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER or doctor/physician.

P363 Wash contaminated clothing before reuse.

Storage

P405 Store locked up.

Disposal

P501 Dispose of contents/container in accordance with local/national regulations

Product label will display most significant precautionary statements.

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Section 2 Hazards identification (Continued)

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

See Section 11 Toxicological Information for more detailed health information.

Section 3 Composition and information on ingredients

3.2 Mixtures

Hazardous ingredients:		Hazard classification of pure ingredients		
Chemical name	% by wt.	EU 1272/2008 CLP/GHS	GHS	Note
Sulfuric Acid CAS # 7664-93-9 EINECS # 231-639-5 Index # 016-020-00-8	20 - 50	Eye Dam. 1, H318 Skin Corr. 1A, H314 Specific Concentration Limit (SCL) Skin Irrit. 2 H315 >= 5% - < 15% Skin Corr. 1A H314 >= 15% Eye Irrit. 2 H319 >= 5% - < 15%	Eye Dam. 1, H318 Skin Corr. 1A, H314	REST
Hydrochloric Acid CAS # 7647-01-0 EINECS # 231-595-7 Index # 017-002-00-2	2 - 5	STOT SE 3, H335 Skin Corr. 1B, H314 Specific Concentration Limit (SCL) Skin Irrit. 2 H315 >= 10% - < 25% STOT SE 3 H335 >= 10% Skin Corr. 1B H314 >= 25% Eye Irrit. 2 H319 >= 10% - < 25%	STOT SE 3, H335 Skin Corr. 1B, H314	REST
Sulfosalicylic Acid, Dihydrate CAS # 5965-83-3 EINECS # Not available Index # Not available	0.5 - 1	Acute Tox. Oral 4, H302 Eye Dam. 1, H318 Skin Corr. 1B, H314 Acute Toxicity Estimates (ATE) ATE Oral = 500 mg/kg	Acute Tox. Oral 4, H302 Eye Dam. 1, H318 Skin Corr. 1B, H314	

REST - Subject to Restriction as per Annex XVII of REACH Regulation (EC) No 1907/2006

See section 8 for available Occupational exposure limits

See Section 15 for additional regulatory information

See Section 16 for description of hazard class and hazard statements

Section 4 First aid measures

4.1 Description of first aid measures

Inhalation

If product is inhaled, move exposed individual to fresh air. If individual is not breathing, begin artificial respiration by trained personnel and obtain medical attention immediately.

Eye contact

If product enters eyes, rinse eyes gently with water for 15 minutes or longer, making sure that the eyelid is held open. Obtain medical advice/attention.

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Section 4 First aid measures (Continued)

Skin contact In case of skin contact, rinse with plenty of water for at least 15 minutes. Remove contaminated clothing and shoes. Obtain medical advice/attention.

Ingestion If product is ingested, rinse mouth with water. If irritation or discomfort occurs, obtain medical attention immediately.

4.2 Most important symptoms and effects, both acute and delayed

Corrosive. May cause burns to eyes, skin and respiratory tract.

See Section 11 Toxicological Information for more detailed health information.

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available. Refer to Section 4.1.

Section 5 Firefighting measures

5.1 Extinguishing media In case of fire use carbon dioxide (CO₂), dry chemical, water spray or foam. For large fires use extinguishing media suitable for surrounding fire.

5.2 Special hazards arising from the substance or mixture
Special fire and explosion hazards

No special hazards determined.

Hazardous combustion products

No combustion products posing significant hazards are expected from this product.

5.3 Advice for firefighters

Protective equipment Self-contained breathing apparatus is recommended for firefighters in all chemical fire situations.

Additional information No further relevant information available.

Section 6 Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions Wear protective gloves, protective clothing and tightly sealed eye/face protection. Observe general safety guidelines for protection; avoid eye and skin contact.

6.2 Environmental precautions Contain spill to prevent migration.
Do not allow the undiluted product to enter sewers/surface or ground water.

6.3 Methods and material for containment and cleaning up

Spill and leak procedures Absorb spilled material with an appropriate inert, non-flammable absorbent and dispose according to local regulations.

6.4 Reference to other sections Refer sections 8 and 13.

Section 7 Handling and storage

7.1 Precautions for safe handling Use good laboratory procedures; avoid eye and skin contact.

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Section 7 Handling and storage (Continued)

7.2 Conditions for safe storage, including any incompatibilities

Store at 2 to 8°C, as directed on the product label.

To maintain product quality, store according to the instructions in the product labeling.

Store away from strong acids, strong bases, strong oxidizers and incompatible materials (section 10).

7.3 Specific end uses

No further relevant information available.

Section 8 Exposure controls and personal protection

8.1 Control parameters

Exposure limits

US OSHA

Sulfuric Acid
CAS # 7664-93-9 1 mg/m³ TWA

Hydrochloric Acid
CAS # 7647-01-0 5 ppm Ceiling; 7 mg/m³ Ceiling

ACGIH

Sulfuric Acid
CAS # 7664-93-9 0.2 mg/m³ TWA (thoracic particulate matter)

Hydrochloric Acid
CAS # 7647-01-0 2 ppm Ceiling

ACGIH Biological Exposure Indices (BEI)

None established

DFG MAK

Sulfuric Acid
CAS # 7664-93-9 0.1 mg/m³ Peak (a momentary value of 0.2 mg/m³ must not be exceeded) (inhalable fraction); 0.1 mg/m³ TWA MAK (inhalable fraction)

Hydrochloric Acid
CAS # 7647-01-0 4 ppm Peak; 6 mg/m³ Peak; 2 ppm TWA MAK; 3.0 mg/m³ TWA MAK

Ireland

Sulfuric Acid
CAS # 7664-93-9 0.05 ppm TWA; 0.15 ppm STEL (calculated)

Hydrochloric Acid
CAS # 7647-01-0 8 mg/m³ TWA (as F); 5 ppm TWA; 10 ppm STEL; 15 mg/m³ STEL

IOELVs

Sulfuric Acid
CAS # 7664-93-9 0.05 mg/m³ TWA

Hydrochloric Acid
CAS # 7647-01-0 5 ppm TWA; 8 mg/m³ TWA; 10 ppm STEL; 15 mg/m³ STEL

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Section 8 Exposure controls and personal protection (Continued)

NIOSH

Sulfuric Acid
CAS # 7664-93-9 15 mg/m3 IDLH; 1 mg/m3 TWA

Hydrochloric Acid
CAS # 7647-01-0 50 ppm IDLH

China

Sulfuric Acid
CAS # 7664-93-9 2 mg/m3 STEL; 1 mg/m3 TWA

Hydrochloric Acid
CAS # 7647-01-0 7.5 mg/m3 Ceiling MAC

Croatia

Sulfuric Acid
CAS # 7664-93-9 0.05 mg/m3 TWA [GVI] (when selecting the appropriate exposure monitoring method the potential limitations and disturbances that may occur in the presence of other sulfur compounds should be taken into account) (fog, thoracic fraction)

Hydrochloric Acid
CAS # 7647-01-0 5 ppm TWA [GVI]; 8 mg/m3 TWA [GVI]; 10 ppm STEL [KGVI]; 15 mg/m3 STEL [KGVI]

Japan

None established

Sweden (AFS 2015:7 and amendments)

Sulfuric Acid
CAS # 7664-93-9 0.1 mg/m3 TLV NGV; 0.2 mg/m3 Indicative STEL Vägledande KGV

Hydrochloric Acid
CAS # 7647-01-0 2 ppm TLV NGV; 3 mg/m3 TLV NGV; 4 ppm Binding STEL Bindande KGV; 6 mg/m3 Binding STEL Bindande KGV

Turkey

Sulfuric Acid
CAS # 7664-93-9 0.05 mg/m3 TWA (steam)

Hydrochloric Acid
CAS # 7647-01-0 10 ppm STEL; 15 mg/m3 STEL; 5 ppm TWA; 8 mg/m3 TWA

8.2 Exposure controls

Engineering controls

No special engineering controls are required. Use with good general ventilation.

Eye protection

Safety glasses or chemical goggles should be worn to prevent eye contact.

Refer U.S. OSHA 29 CFR 1910.133, European Standard EN166 or appropriate government standards.

Skin protection

Wear impervious gloves such as Nitrile or equivalent and protective clothing. Refer to U.S. OSHA 29 CFR 1910.138, European Standard EN 374, EN 14605:2005+A1:2009 or appropriate government standards.

Respiratory protection

Under normal conditions, the use of this product should not require respiratory protection. If overexposure should occur and ventilation is not adequate to maintain airborne concentrations at acceptable levels, the use of respiratory protection should be evaluated by a qualified professional.

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Section 9 Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid	Density and/or relative density	1.141
Color	Colorless	Solubility	
Odor	Characteristic odor	Water	Miscible
pH	< 1	Organic	Not determined
Freezing point	Similar to water, approximately 0°C	Partition coefficient n-octanol/water (log value)	Not determined
Boiling point or initial boiling point and boiling range	Similar to water, approximately 100°C	Auto-ignition temp.	Product is not selfigniting
Flash point	Not applicable	Decomposition temperature	Not determined
Flammability	Not applicable	Vapor pressure	Similar to water, approximately 23 hPa
		Kinematic viscosity	Not determined
Lower and upper explosion limit	Not applicable		
Relative vapor density	Not determined		
Particle characteristics	Not applicable		

9.2 Other information

Information with regard to physical hazard classes

Not applicable
No further relevant information available.

Other safety characteristics

No further relevant information available.

Section 10 Stability and reactivity

10.1 Reactivity	No further relevant information available.
10.2 Chemical stability	The product is stable in accordance with recommended storage conditions.
10.3 Possibility of hazardous reactions	No further relevant information available.
10.4 Conditions to avoid	To maintain product performance keep away from strong acids, strong bases, strong oxidizers. Avoid exposure to heat and direct sunlight.

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Section 10 Stability and reactivity (Continued)

10.5 Incompatible materials No further relevant information available.

10.6 Hazardous decomposition products

When stored as labeled, no known hazardous decomposition products are formed during the shelf-life of this product.

Section 11 Toxicological information

11.1 Information on hazard classes

Toxicity data for hazardous ingredients

Sulfuric Acid
CAS # 7664-93-9
Inhalation LC50 Rat 0.375 mg/L 4 h (aerosol)(OECD_SIDS); Oral LD50 Rat 2140 mg/kg (JAPAN_GHS)

Hydrochloric Acid
CAS # 7647-01-0
Dermal LD50 Rabbit >5010 mg/kg (JAPAN_GHS); Inhalation LC50 Rat 1.68 mg/L 1 h (mist)(JAPAN_GHS); Oral LD50 Rat 238 - 277 mg/kg (JAPAN_GHS)

Primary routes of exposure Eye contact, ingestion, inhalation, and skin contact.

Acute toxicity Not classified based on available data.

Skin corrosion/irritation Contact may cause severe skin burns.

Serious eye damage/irritation Contact may cause serious eye damage.

Respiratory or skin sensitisation Not classified based on available data.

Germ cell mutagenicity Not classified based on available data.

Carcinogenicity Sulfuric Acid is listed with OSHA as a carcinogen; IARC as Group 1 carcinogen to humans; ACGIH as A2 Suspected Human Carcinogen.

Reproductive toxicity Not classified based on available data.

Specific target organ toxicity (STOT) – single exposure

Not classified based on available data.

Specific target organ toxicity (STOT) – repeated exposure

Not classified based on available data.

Aspiration hazard Not classified based on available data.

11.2 Information on other hazards

Endocrine disrupting properties

This product does not have substance(s) of endocrine disrupting properties for health according to REACH Article 57(f).

Other information No further relevant information available.

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Section 12 Ecological information

12.1 Toxicity

Fresh water species

Sulfuric Acid
CAS # 7664-93-9

LC50 96 h Brachydanio rerio: >500 mg/L [static] (IUCRID)

Microtox/organisms

No information available.

Water flea

No information available.

Fresh water algae

No information available.

12.2 Persistence and degradability

Not determined for the product.

12.3 Bioaccumulative potential

Not determined for the product.

12.4 Mobility in soil

Not determined for the product.

12.5 Results of PBT and vPvB assessment

Not determined for the product. PBT: Not applicable, vPvB: Not applicable.

12.6 Endocrine disrupting properties

This product does not have substance(s) of endocrine disrupting properties for environment according to REACH Article 57(f).

12.7 Other adverse effects

No further relevant information available.

Section 13 Disposal considerations

13.1 Waste treatment methods

Product waste disposal

Chemical residues and remains should be routinely handled as special waste. This must be disposed of in compliance with anti-pollution and other laws of the country concerned. To ensure compliance we recommend that you contact the relevant (local) authorities and/or an approved waste-disposal company for information.

Package disposal

Dispose of waste product, unused product and contaminated packaging in compliance with federal, state and local regulations. If unsure of the applicable requirements, contact the authorities for information.

Additional information

Suggested European waste catalogue 18 01 06* - chemicals consisting of or containing dangerous substances. Dispose in accordance with national, state and local waste regulations.

Section 14 Transport information

Shipping information	IATA	IMDG	US DOT	European ADR	Canadian TDG
14.1 UN/ID number	3264	3264	3264	3264	PIN - 3264
14.2 UN proper shipping name	Corrosive liquid, acidic, inorganic, n.o.s. (sulphuric acid, hydrochloric acid)				
14.3 Transport hazard class(es)	8 Corrosives	8 Corrosive substances	8 Corrosive material	8 Corrosive substances	8 Corrosives
Subsidiary risk	None	None	None	None	None

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Section 14 Transport information (Continued)

Classification code	Not applicable	Not applicable	Not applicable	C1	Not applicable
14.4 Packing group	II	II	II	II	II
Special provisions	A3, A803	274	386	274	16
Additional information					
IATA ERG code	8L	Not applicable	Not applicable	Not applicable	Not applicable
EmS	Not applicable	F-A, S-B	Not applicable	Not applicable	Not applicable
NAERG code	Not applicable	Not applicable	154	Not applicable	154
14.5 Environmental hazards					
Marine pollutant	Not applicable	No	Not applicable	Not applicable	Not applicable
14.6 Special precautions for user	Warning: Corrosive substances.				
14.7 Maritime transport in bulk according to IMO instruments	Not applicable				

Section 15 Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

US Federal and State Regulations

SARA 313 (Section 313, Title III reporting requirements)

CAS # 7664-93-9	Sulfuric Acid	1.0% de minimis concentration
CAS # 7647-01-0	Hydrochloric Acid	1.0% de minimis concentration

CERCLA (The Comprehensive Environmental Response, Compensation, and Liability Act) 40 CFR 302.4

CAS # 7664-93-9	Sulfuric Acid
CAS # 7647-01-0	Hydrochloric Acid

California Proposition 65

⚠ WARNING This product can expose you to chemical which is known to the State of California to cause cancer and/or reproductive harm. For more information go to www.P65Warnings.ca.gov

Chemical which is known to the State of California to cause cancer

CAS # 7664-93-9	Sulfuric Acid
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Chemical which is known to the State of California to cause development toxicity

No ingredients listed.

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Section 15 Regulatory information (Continued)

Chemical which is known to the State of California to cause male reproductive toxicity

No ingredients listed.

Chemical which is known to the State of California to cause female reproductive toxicity

No ingredients listed.

Massachusetts Right To Know (RTK) List

CAS # 7664-93-9	Sulfuric Acid
CAS # 7647-01-0	Hydrochloric Acid

New Jersey Dept. of Health Right To Know (RTK) List

CAS # 7664-93-9	Sulfuric Acid
CAS # 7647-01-0	Hydrochloric Acid

Pennsylvania Right To Know (RTK) List

CAS # 7664-93-9	Sulfuric Acid
CAS # 7647-01-0	Hydrochloric Acid

EU Regulations

This SDS complies with EC Regulations 1907/2006 (REACH) and amendments.

Water Hazard Class (Germany)

WGK 2, water endangering

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors - Substances Subject to Suspicious Transactions Reporting

CAS # 7664-93-9	Sulfuric Acid
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This product is regulated by Regulation (EU) 2019/1148: all suspicious transactions, and significant disappearances and thefts should be reported to the relevant national contact point.

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors - Restricted Explosives Precursors

CAS # 7664-93-9	Sulfuric Acid
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Acquisition, introduction, possession, or use of this product by the general public is restricted by Regulation (EU) 2019/1148. All suspicious transactions, and significant disappearances and thefts should be reported to the relevant national contact point

REACH 1907/2006 EC - Candidate List of Substances of Very High Concern (SVHC)

No ingredients listed.

REACH 1907/2006 EC - Annex XVII – Restrictions on Certain Dangerous Substances

CAS # 7664-93-9	Sulfuric Acid	Entry No.: 75 (B)
CAS # 7647-01-0	Hydrochloric Acid	Entry No.: 75 (U,5)

REACH 1907/2006 EC - Annex XIV - list of substances subject to authorisation

No ingredients listed.

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Section 15 Regulatory information (Continued)

UK Regulations

UK REACH Regulation (as Amended) - List of substances subject to authorisation

Refer to Section 3

Canada

This product is exempt from WHMIS label and SDS requirements.

China

Catalog of Hazardous Chemicals - Hazardous Chemicals

CAS # 7664-93-9 Sulfuric Acid
CAS # 7647-01-0 Hydrochloric Acid

Inventory - China - Inventory of Existing Chemical Substances (IECSC)

All ingredients are listed or exempted.

Turkey

Türkiye-REACH - KKDIK Regulation - Annex 17 – Restrictions

No ingredients listed.

International

UN/FAO/Rotterdam Convention - Chemicals Subject to Prior Informed Consent (PIC)

No ingredients listed.

15.2 Chemical safety assessment

A Chemical Safety Assessment has not been carried out.

Some hazardous ingredients listed in Section 15 are below the cutoff limits of 0.1% for carcinogen, mutagen and reproductive toxin and 1% for other health hazards required for reporting in Section 3.

Section 16 Other information

Beckman Coulter safety rating	Flammability: 0 Health: 3 Reactivity with water: 0 Physical contact: 3	Code 0=None 1=Slight 2=Caution 3=Severe
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Revision changes

Update supplier addresses in Section 1.3

Document version and issue/revision date

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Section 16 Other information (Continued)

Hazard Classification Procedure This mixture was classified using the calculation method for human health and environmental hazards. Physical hazards were determined based on the specification.

Description of hazard class and hazard statements from Section 3

Acute Tox. Oral 4 - Acute Toxicity Oral, Category 4
Eye Dam. 1 - Eye Damage Category 1
Skin Corr. 1A - Skin Corrosion Category 1A
Skin Corr. 1B - Skin Corrosion Category 1B
STOT SE 3 - Specific Target Organ Toxicity Single Exposure Category 3
H302 - Harmful if swallowed.
H314 - Causes severe skin burns and eye damage.
H318 - Causes serious eye damage.
H335 - May cause respiratory irritation.

Abbreviations and acronyms

ACGIH - American Conference of Governmental Industrial Hygienists (ACGIH)
ADR and RID - European Agreement Concerning the International Carriage of Dangerous Goods by Road and Rail
CLP - Classification, Labeling and Packaging
DFGMAK - Republic Germany's maximum exposure limit
EC50 - Concentration of a substance in an environmental medium expected to produce a certain effect in 50% of test organisms
GHS - Globally Harmonized System of Classification and Labelling of Chemicals (GHS)
HCS - Hazard Communication Standard
IARC - International Agency for Research on Cancer
IATA DGR - International Air Transport Association Dangerous Goods Regulation
ICAO - International Civil Aviation Organization
IDLH - Immediately Dangerous to Life or Health
IMDG - International Maritime Dangerous Goods
IMO - International Maritime Organization
IOELVs - European Unions' Indicative Occupational Exposure Limit Values
LC50 - Concentration of a substance in water causing death (50% of the tested population) to aquatic life
LD50 - Lethal Dose 50%
NIOSH - National Institute for Occupational Safety and Health
NTP - National Toxicology Program
OSHA - Occupational Safety and Health Administration
PBT - Persistent Bioaccumulative and Toxic substances
PEL - Permissible Exposure Limit
SARA - Superfund Amendments and Reauthorization Act
STEL – Short Term Exposure Limit
STLV - Short Term Limit Value

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Section 16 Other information (Continued)

STV - Short Term Value
TDG - Canadian Transportation of Dangerous Goods Regulations
TLV - Threshold Limit Value
TWA – Time Weighted Average
UN GHS - United Nations Globally Harmonized System
US DOT - United States Department of Transportation
US OSHA - United States Occupational Safety and Health Administration
vPvB - very Persistent and very Bioaccumulative substances
WHMIS - Workplace Hazardous Material Information System

For further information, please contact your local Beckman Coulter, Inc. representative.

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Section 1 Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name Direct Bilirubin Color R1
Part number Component of P/N OSR6111, OSR6211, OSR6611

1.2 Relevant identified uses of the substance or mixture and uses advised against

Product use For In Vitro Diagnostic Use. See product literature for details.
 For Professional Use Only

1.3 Details of the supplier of the safety data sheet

Manufacturer

Beckman Coulter, Inc.
 250 S. Kraemer Blvd
 Brea, CA 92821, U.S.A.
 Tel: 800-854-3633

Supplier

CANADA
 Beckman Coulter Canada LP
 7075 Financial Drive
 Mississauga, ON L5N 6V8
 Canada
 1-800-463-7828

UNITED KINGDOM
 Beckman Coulter (UK) Ltd.
 Amersham Place
 Little Chalfont
 Buckinghamshire
 United Kingdom, HP7 9NA
 01494 441181

AUSTRALIA
 Beckman Coulter Australia Pty Ltd
 23-27 Chaplin Drive
 Lane Cove NSW 2066
 Australia
 ABN 81 002 011 672
 24 Hour emergency contact phone
 number:
 1800 060 881

SWITZERLAND
 Beckman Coulter Eurocenter SA
 22, rue Juste-Olivier, Case Postale
 1044,
 CH-1260 Nyon 1, Switzerland.
 Telephone: +41 (0)22 365 36 11
 Monday through Friday, 9:00 am to
 7:00pm)

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Section 1 Identification of the substance/mixture and of the company/undertaking (Continued)

NEW ZEALAND
Beckman Coulter NZ
Unit J, 33 Walmsley Road, Otahuhu,
Auckland 1062, New Zealand
Hours available: 08:30 - 17:00

Beckman Coulter Ireland Inc.
Lismeehan
O'Callaghan's Mills
Co. Clare
Ireland
Tel: 353 (0)65 6831100

ICELAND / ÍSLAND
Beckman Coulter AB
Ekbacksvägen 28
168 69 Bromma
Sweden
Phone No.: +46 80564 85 900
Hours available: 08.00-16.30

MALTA
DX Distributor:
Cherubino Ltd
DELFI Building, Sliema Road, Gzira,
GZR 1637
Telephone: +356 21343270
Hours available: 08:30 – 17:00

e-mail address

SDSNT@beckman.com
Further information Contact:
Customer support Unit, Beckman Coulter Ireland Inc.
Technical Service Department Tel. +001-800-854-3633 (PST)
E-mail: Techsupportuk@beckman.com

1.4 Emergency telephone number

Telephone number (24H)

Chemtrec Emergency Tel No. U.S.A. 800-424-9300, International (001) 703-527-3887
Tel +353 (0)65 683 1170; 08:00 - 16:30 hrs Mon-Thur, 08:00 - 15:30 hrs Fri (GMT) Tel +001-800-223-0130 (PST)

Distributor and emergency phone no.

Refer to attached list, Document ID: [472050](#), for local distributor and emergency phone numbers.

UNITED STATES - Emergency Phone (24h): Chemtrec (800) 424-9300, International (001) 703-527-3887

CANADA - Poison Centre: 1-844-764-7669; Centre antipoison du Québec: 1-800-463-5060

UNITED KINGDOM - For UK and Scotland: Emergency Call 999

IRELAND - National Poisons Information Centre Phone No.: Members of Public: +353 (01) 809 2166 (8:00 am to 10:00 pm 7 days a week); Healthcare Professionals: +353 (01) 809 2566 (24 hour service)

AUSTRALIA - 24 Hour emergency contact phone number: 1800 060 881

NEW ZEALAND - 24 Hour emergency number: 0800 446 109

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Section 1 Identification of the substance/mixture and of the company/undertaking (Continued)

Section 2 Hazards identification

2.1 Classification of the substance or mixture

Product description In vitro diagnostic reagent.

Colorless; Liquid; Odorless

Classification according to EC 1272/2008 (CLP/GHS)

Skin Corrosion Category 1, H314

Eye Damage Category 1, H318

Classification according to US-OSHA (HCS 29 CFR 1910.1200) and UN GHS

Skin Corrosion Category 1

Eye Damage Category 1

2.2 Label elements

According to EC 1272/2008 (CLP/GHS), US-OSHA and UN GHS

Hazardous ingredients

Sulfuric Acid

Sulfosalicylic Acid, Dihydrate

Hydrochloric Acid

Pictogram



Signal word

DANGER

Hazard statements

H314 Causes severe skin burns and eye damage.

Precautionary statements

Prevention

P280 Wear protective gloves, protective clothing and eye/face protection.

Response

P301+P330+P331 IF SWALLOWED: rinse mouth. Do NOT induce vomiting.

P303+P361+P353 IF ON SKIN (or hair): Rinse skin with water.

P304+P340 IF INHALED: Remove person to fresh air and keep at rest in a position comfortable for breathing.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER or doctor/physician.

P363 Wash contaminated clothing before reuse.

Storage

P405 Store locked up.

Disposal

P501 Dispose of contents/container in accordance with local/national regulations

Product label will display most significant precautionary statements.

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Section 2 Hazards identification (Continued)

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

See Section 11 Toxicological Information for more detailed health information.

Section 3 Composition and information on ingredients

3.2 Mixtures

Hazardous ingredients:		Hazard classification of pure ingredients		
Chemical name	% by wt.	EU 1272/2008 CLP/GHS	GHS	Note
Sulfuric Acid CAS # 7664-93-9 EINECS # 231-639-5 Index # 016-020-00-8	20 - 50	Eye Dam. 1, H318 Skin Corr. 1A, H314 Specific Concentration Limit (SCL) Skin Irrit. 2 H315 >= 5% - < 15% Skin Corr. 1A H314 >= 15% Eye Irrit. 2 H319 >= 5% - < 15%	Eye Dam. 1, H318 Skin Corr. 1A, H314	REST
Hydrochloric Acid CAS # 7647-01-0 EINECS # 231-595-7 Index # 017-002-00-2	2 - 5	STOT SE 3, H335 Skin Corr. 1B, H314 Specific Concentration Limit (SCL) Skin Irrit. 2 H315 >= 10% - < 25% STOT SE 3 H335 >= 10% Skin Corr. 1B H314 >= 25% Eye Irrit. 2 H319 >= 10% - < 25%	STOT SE 3, H335 Skin Corr. 1B, H314	REST
Sulfosalicylic Acid, Dihydrate CAS # 5965-83-3 EINECS # Not available Index # Not available	0.5 - 1	Acute Tox. Oral 4, H302 Eye Dam. 1, H318 Skin Corr. 1B, H314 Acute Toxicity Estimates (ATE) ATE Oral = 500 mg/kg	Acute Tox. Oral 4, H302 Eye Dam. 1, H318 Skin Corr. 1B, H314	

REST - Subject to Restriction as per Annex XVII of REACH Regulation (EC) No 1907/2006

See section 8 for available Occupational exposure limits

See Section 15 for additional regulatory information

See Section 16 for description of hazard class and hazard statements

Section 4 First aid measures

4.1 Description of first aid measures

Inhalation

If product is inhaled, move exposed individual to fresh air. If individual is not breathing, begin artificial respiration by trained personnel and obtain medical attention immediately.

Eye contact

If product enters eyes, rinse eyes gently with water for 15 minutes or longer, making sure that the eyelid is held open. Obtain medical advice/attention.

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Section 4 First aid measures (Continued)

Skin contact In case of skin contact, rinse with plenty of water for at least 15 minutes. Remove contaminated clothing and shoes. Obtain medical advice/attention.

Ingestion If product is ingested, rinse mouth with water. If irritation or discomfort occurs, obtain medical attention immediately.

4.2 Most important symptoms and effects, both acute and delayed

Corrosive. May cause burns to eyes, skin and respiratory tract.
See Section 11 Toxicological Information for more detailed health information.

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available. Refer to Section 4.1.

Section 5 Firefighting measures

5.1 Extinguishing media In case of fire use carbon dioxide (CO₂), dry chemical, water spray or foam.
For large fires use extinguishing media suitable for surrounding fire.

5.2 Special hazards arising from the substance or mixture
Special fire and explosion hazards

No special hazards determined.

Hazardous combustion products

No combustion products posing significant hazards are expected from this product.

5.3 Advice for firefighters

Protective equipment Self-contained breathing apparatus is recommended for firefighters in all chemical fire situations.

Additional information No further relevant information available.

Section 6 Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions Wear protective gloves, protective clothing and tightly sealed eye/face protection.
Observe general safety guidelines for protection; avoid eye and skin contact.

6.2 Environmental precautions Contain spill to prevent migration.
Do not allow the undiluted product to enter sewers/surface or ground water.

6.3 Methods and material for containment and cleaning up

Spill and leak procedures Absorb spilled material with an appropriate inert, non-flammable absorbent and dispose according to local regulations.

6.4 Reference to other sections Refer sections 8 and 13.

Section 7 Handling and storage

7.1 Precautions for safe handling Use good laboratory procedures; avoid eye and skin contact.

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Section 7 Handling and storage (Continued)

7.2 Conditions for safe storage, including any incompatibilities

Store at 2 to 8°C, as directed on the product label.

To maintain product quality, store according to the instructions in the product labeling.

Store away from strong acids, strong bases, strong oxidizers and incompatible materials (section 10).

7.3 Specific end uses

No further relevant information available.

Section 8 Exposure controls and personal protection

8.1 Control parameters

Exposure limits

US OSHA

Sulfuric Acid
CAS # 7664-93-9 1 mg/m³ TWA

Hydrochloric Acid
CAS # 7647-01-0 5 ppm Ceiling; 7 mg/m³ Ceiling

ACGIH

Sulfuric Acid
CAS # 7664-93-9 0.2 mg/m³ TWA (thoracic particulate matter)

Hydrochloric Acid
CAS # 7647-01-0 2 ppm Ceiling

ACGIH Biological Exposure Indices (BEI)

None established

DFG MAK

Sulfuric Acid
CAS # 7664-93-9 0.1 mg/m³ Peak (a momentary value of 0.2 mg/m³ must not be exceeded) (inhalable fraction); 0.1 mg/m³ TWA MAK (inhalable fraction)

Hydrochloric Acid
CAS # 7647-01-0 4 ppm Peak; 6 mg/m³ Peak; 2 ppm TWA MAK; 3.0 mg/m³ TWA MAK

Ireland

Sulfuric Acid
CAS # 7664-93-9 0.05 ppm TWA; 0.15 ppm STEL (calculated)

Hydrochloric Acid
CAS # 7647-01-0 8 mg/m³ TWA (as F); 5 ppm TWA; 10 ppm STEL; 15 mg/m³ STEL

IOELVs

Sulfuric Acid
CAS # 7664-93-9 0.05 mg/m³ TWA

Hydrochloric Acid
CAS # 7647-01-0 5 ppm TWA; 8 mg/m³ TWA; 10 ppm STEL; 15 mg/m³ STEL

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Section 8 Exposure controls and personal protection (Continued)

NIOSH

Sulfuric Acid
CAS # 7664-93-9 15 mg/m3 IDLH; 1 mg/m3 TWA

Hydrochloric Acid
CAS # 7647-01-0 50 ppm IDLH

China

Sulfuric Acid
CAS # 7664-93-9 2 mg/m3 STEL; 1 mg/m3 TWA

Hydrochloric Acid
CAS # 7647-01-0 7.5 mg/m3 Ceiling MAC

Croatia

Sulfuric Acid
CAS # 7664-93-9 0.05 mg/m3 TWA [GVI] (when selecting the appropriate exposure monitoring method the potential limitations and disturbances that may occur in the presence of other sulfur compounds should be taken into account) (fog, thoracic fraction)

Hydrochloric Acid
CAS # 7647-01-0 5 ppm TWA [GVI]; 8 mg/m3 TWA [GVI]; 10 ppm STEL [KGVI]; 15 mg/m3 STEL [KGVI]

Japan

None established

Sweden (AFS 2015:7 and amendments)

Sulfuric Acid
CAS # 7664-93-9 0.1 mg/m3 TLV NGV; 0.2 mg/m3 Indicative STEL Vägledande KGV

Hydrochloric Acid
CAS # 7647-01-0 2 ppm TLV NGV; 3 mg/m3 TLV NGV; 4 ppm Binding STEL Bindande KGV; 6 mg/m3 Binding STEL Bindande KGV

Turkey

Sulfuric Acid
CAS # 7664-93-9 0.05 mg/m3 TWA (steam)

Hydrochloric Acid
CAS # 7647-01-0 10 ppm STEL; 15 mg/m3 STEL; 5 ppm TWA; 8 mg/m3 TWA

8.2 Exposure controls

Engineering controls

No special engineering controls are required. Use with good general ventilation.

Eye protection

Safety glasses or chemical goggles should be worn to prevent eye contact.

Refer U.S. OSHA 29 CFR 1910.133, European Standard EN166 or appropriate government standards.

Skin protection

Wear impervious gloves such as Nitrile or equivalent and protective clothing. Refer to U.S. OSHA 29 CFR 1910.138, European Standard EN 374, EN 14605:2005+A1:2009 or appropriate government standards.

Respiratory protection

Under normal conditions, the use of this product should not require respiratory protection. If overexposure should occur and ventilation is not adequate to maintain airborne concentrations at acceptable levels, the use of respiratory protection should be evaluated by a qualified professional.

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Section 9 Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid	Density and/or relative density	1.14
Color	Colorless	Solubility	
Odor	Odorless	Water	Not miscible
pH	< 1	Organic	Not determined
Freezing point	Similar to water, approximately 0°C	Partition coefficient n-octanol/water (log value)	Not determined
Boiling point or initial boiling point and boiling range	Similar to water, approximately 100°C	Auto-ignition temp.	Product is not selfigniting
Flash point	Not applicable	Decomposition temperature	Not determined
Flammability	Not applicable	Vapor pressure	Similar to water, approximately 23 hPa
		Kinematic viscosity	Not determined
Lower and upper explosion limit	Not applicable		
Relative vapor density	Not determined		
Particle characteristics	Not applicable		

9.2 Other information

Information with regard to physical hazard classes

Not applicable
No further relevant information available.

Other safety characteristics

No further relevant information available.

Section 10 Stability and reactivity

10.1 Reactivity	No further relevant information available.
10.2 Chemical stability	The product is stable in accordance with recommended storage conditions.
10.3 Possibility of hazardous reactions	No further relevant information available.
10.4 Conditions to avoid	To maintain product performance keep away from strong acids, strong bases, strong oxidizers. Avoid exposure to heat and direct sunlight.

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Section 10 Stability and reactivity (Continued)

10.5 Incompatible materials No further relevant information available.

10.6 Hazardous decomposition products

When stored as labeled, no known hazardous decomposition products are formed during the shelf-life of this product.

Section 11 Toxicological information

11.1 Information on hazard classes

Toxicity data for hazardous ingredients

Sulfuric Acid
CAS # 7664-93-9
Inhalation LC50 Rat 0.375 mg/L 4 h (aerosol)(OECD_SIDS); Oral LD50 Rat 2140 mg/kg (JAPAN_GHS)

Hydrochloric Acid
CAS # 7647-01-0
Dermal LD50 Rabbit >5010 mg/kg (JAPAN_GHS); Inhalation LC50 Rat 1.68 mg/L 1 h (mist)(JAPAN_GHS); Oral LD50 Rat 238 - 277 mg/kg (JAPAN_GHS)

Primary routes of exposure Eye contact, ingestion, inhalation, and skin contact.

Acute toxicity Not classified based on available data.

Skin corrosion/irritation Contact may cause severe skin burns.

Serious eye damage/irritation Contact may cause serious eye damage.

Respiratory or skin sensitisation Not classified based on available data.

Germ cell mutagenicity Not classified based on available data.

Carcinogenicity Sulfuric Acid is listed with OSHA as a carcinogen; IARC as Group 1 carcinogen to humans; ACGIH as A2 Suspected Human Carcinogen.

Reproductive toxicity Not classified based on available data.

Specific target organ toxicity (STOT) – single exposure

Not classified based on available data.

Specific target organ toxicity (STOT) – repeated exposure

Not classified based on available data.

Aspiration hazard Not classified based on available data.

11.2 Information on other hazards

Endocrine disrupting properties

This product does not have substance(s) of endocrine disrupting properties for health according to REACH Article 57(f).

Other information No further relevant information available.

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Section 12 Ecological information

12.1 Toxicity

Fresh water species

Sulfuric Acid
CAS # 7664-93-9

LC50 96 h Brachydanio rerio: >500 mg/L [static] (IUCRID)

Microtox/organisms

No information available.

Water flea

No information available.

Fresh water algae

No information available.

12.2 Persistence and degradability

Not determined for the product.

12.3 Bioaccumulative potential

Not determined for the product.

12.4 Mobility in soil

Not determined for the product.

12.5 Results of PBT and vPvB assessment

Not determined for the product. PBT: Not applicable, vPvB: Not applicable.

12.6 Endocrine disrupting properties

This product does not have substance(s) of endocrine disrupting properties for environment according to REACH Article 57(f).

12.7 Other adverse effects

No further relevant information available.

Section 13 Disposal considerations

13.1 Waste treatment methods

Product waste disposal

Chemical residues and remains should be routinely handled as special waste. This must be disposed of in compliance with anti-pollution and other laws of the country concerned. To ensure compliance we recommend that you contact the relevant (local) authorities and/or an approved waste-disposal company for information.

Package disposal

Dispose of waste product, unused product and contaminated packaging in compliance with federal, state and local regulations. If unsure of the applicable requirements, contact the authorities for information.

Additional information

Suggested European waste catalogue 18 01 06* - chemicals consisting of or containing dangerous substances. Dispose in accordance with national, state and local waste regulations.

Section 14 Transport information

Shipping information	IATA	IMDG	US DOT	European ADR	Canadian TDG
14.1 UN/ID number	3264	3264	3264	3264	PIN - 3264
14.2 UN proper shipping name	Corrosive liquid, acidic, inorganic, n.o.s. (sulphuric acid, hydrochloric acid)				
14.3 Transport hazard class(es)	8 Corrosives	8 Corrosive substances	8 Corrosive material	8 Corrosive substances	8 Corrosives
Subsidiary risk	None	None	None	None	None

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Section 14 Transport information (Continued)

Classification code	Not applicable	Not applicable	Not applicable	C1	Not applicable
14.4 Packing group	II	II	II	II	II
Special provisions	A3, A803	274	386	274	16
Additional information					
IATA ERG code	8L	Not applicable	Not applicable	Not applicable	Not applicable
EmS	Not applicable	F-A, S-B	Not applicable	Not applicable	Not applicable
NAERG code	Not applicable	Not applicable	154	Not applicable	154
14.5 Environmental hazards					
Marine pollutant	Not applicable	No	Not applicable	Not applicable	Not applicable
14.6 Special precautions for user	Warning: Corrosive substances.				
14.7 Maritime transport in bulk according to IMO instruments	Not applicable				

Section 15 Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

US Federal and State Regulations

SARA 313 (Section 313, Title III reporting requirements)

CAS # 7664-93-9	Sulfuric Acid	1.0% de minimis concentration
CAS # 7647-01-0	Hydrochloric Acid	1.0% de minimis concentration

CERCLA (The Comprehensive Environmental Response, Compensation, and Liability Act) 40 CFR 302.4

CAS # 7664-93-9	Sulfuric Acid
CAS # 7647-01-0	Hydrochloric Acid

California Proposition 65

⚠ WARNING This product can expose you to chemical which is known to the State of California to cause cancer and/or reproductive harm. For more information go to www.P65Warnings.ca.gov

Chemical which is known to the State of California to cause cancer

CAS # 7664-93-9	Sulfuric Acid
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Chemical which is known to the State of California to cause development toxicity

No ingredients listed.

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Section 15 Regulatory information (Continued)

Chemical which is known to the State of California to cause male reproductive toxicity

No ingredients listed.

Chemical which is known to the State of California to cause female reproductive toxicity

No ingredients listed.

Massachusetts Right To Know (RTK) List

CAS # 7664-93-9	Sulfuric Acid
CAS # 7647-01-0	Hydrochloric Acid

New Jersey Dept. of Health Right To Know (RTK) List

CAS # 7664-93-9	Sulfuric Acid
CAS # 7647-01-0	Hydrochloric Acid

Pennsylvania Right To Know (RTK) List

CAS # 7664-93-9	Sulfuric Acid
CAS # 7647-01-0	Hydrochloric Acid

EU Regulations

This SDS complies with EC Regulations 1907/2006 (REACH) and amendments.

Water Hazard Class (Germany)

WGK 1, low water endangering

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors - Substances Subject to Suspicious Transactions Reporting

CAS # 7664-93-9	Sulfuric Acid
-----------------	---------------

This product is regulated by Regulation (EU) 2019/1148: all suspicious transactions, and significant disappearances and thefts should be reported to the relevant national contact point.

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors - Restricted Explosives Precursors

CAS # 7664-93-9	Sulfuric Acid
-----------------	---------------

Acquisition, introduction, possession, or use of this product by the general public is restricted by Regulation (EU) 2019/1148. All suspicious transactions, and significant disappearances and thefts should be reported to the relevant national contact point

REACH 1907/2006 EC - Candidate List of Substances of Very High Concern (SVHC)

No ingredients listed.

REACH 1907/2006 EC - Annex XVII – Restrictions on Certain Dangerous Substances

CAS # 7664-93-9	Sulfuric Acid	Entry No.: 75 (B)
CAS # 7647-01-0	Hydrochloric Acid	Entry No.: 75 (U,5)

REACH 1907/2006 EC - Annex XIV - list of substances subject to authorisation

No ingredients listed.

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Section 15 Regulatory information (Continued)

UK Regulations

UK REACH Regulation (as Amended) - List of substances subject to authorisation

Refer to Section 3

Canada

This product is exempt from WHMIS label and SDS requirements.

China

Catalog of Hazardous Chemicals - Hazardous Chemicals

CAS # 7664-93-9 Sulfuric Acid
CAS # 7647-01-0 Hydrochloric Acid

Inventory - China - Inventory of Existing Chemical Substances (IECSC)

All ingredients are listed or exempted.

Turkey

Türkiye-REACH - KKDIK Regulation - Annex 17 – Restrictions

No ingredients listed.

International

UN/FAO/Rotterdam Convention - Chemicals Subject to Prior Informed Consent (PIC)

No ingredients listed.

15.2 Chemical safety assessment

A Chemical Safety Assessment has not been carried out.

Some hazardous ingredients listed in Section 15 are below the cutoff limits of 0.1% for carcinogen, mutagen and reproductive toxin and 1% for other health hazards required for reporting in Section 3.

Section 16 Other information

Beckman Coulter safety rating	Flammability: 0 Health: 3 Reactivity with water: 0 Physical contact: 3	Code 0=None 1=Slight 2=Caution 3=Severe
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Revision changes Update supplier addresses in Section 1.3

Document version and issue/revision date

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Section 16 Other information (Continued)

Hazard Classification Procedure This mixture was classified using the calculation method for human health and environmental hazards. Physical hazards were determined based on the specification.

Description of hazard class and hazard statements from Section 3

Acute Tox. Oral 4 - Acute Toxicity Oral, Category 4
Eye Dam. 1 - Eye Damage Category 1
Skin Corr. 1A - Skin Corrosion Category 1A
Skin Corr. 1B - Skin Corrosion Category 1B
STOT SE 3 - Specific Target Organ Toxicity Single Exposure Category 3
H302 - Harmful if swallowed.
H314 - Causes severe skin burns and eye damage.
H318 - Causes serious eye damage.
H335 - May cause respiratory irritation.

Abbreviations and acronyms

ACGIH - American Conference of Governmental Industrial Hygienists (ACGIH)
ADR and RID - European Agreement Concerning the International Carriage of Dangerous Goods by Road and Rail
CLP - Classification, Labeling and Packaging
DFGMAK - Republic Germany's maximum exposure limit
EC50 - Concentration of a substance in an environmental medium expected to produce a certain effect in 50% of test organisms
GHS - Globally Harmonized System of Classification and Labelling of Chemicals (GHS)
HCS - Hazard Communication Standard
IARC - International Agency for Research on Cancer
IATA DGR - International Air Transport Association Dangerous Goods Regulation
ICAO - International Civil Aviation Organization
IDLH - Immediately Dangerous to Life or Health
IMDG - International Maritime Dangerous Goods
IMO - International Maritime Organization
IOELVs - European Unions' Indicative Occupational Exposure Limit Values
LC50 - Concentration of a substance in water causing death (50% of the tested population) to aquatic life
LD50 - Lethal Dose 50%
NIOSH - National Institute for Occupational Safety and Health
NTP - National Toxicology Program
OSHA - Occupational Safety and Health Administration
PBT - Persistent Bioaccumulative and Toxic substances
PEL - Permissible Exposure Limit
SARA - Superfund Amendments and Reauthorization Act
STEL – Short Term Exposure Limit
STLV - Short Term Limit Value

SAFETY DATA SHEET

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Section 16 Other information (Continued)

STV - Short Term Value
TDG - Canadian Transportation of Dangerous Goods Regulations
TLV - Threshold Limit Value
TWA – Time Weighted Average
UN GHS - United Nations Globally Harmonized System
US DOT - United States Department of Transportation
US OSHA - United States Occupational Safety and Health Administration
vPvB - very Persistent and very Bioaccumulative substances
WHMIS - Workplace Hazardous Material Information System

For further information, please contact your local Beckman Coulter, Inc. representative.

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